

Sarvajith Kujuluva

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2027

Bachelor of Science in Computer Engineering; GPA: 4.00

Threads: Computer Hardware & Emerging Architecture + Systems & Architecture

EXPERIENCE

SiliconJackets ASIC Design Team, Atlanta, GA

August 2025 – Present

RTL & FSM Design / Hardware Logic

- Built a 64-bit calculator in **SystemVerilog** using cascaded 32-bit adders, a seven-state FSM, and memory-mapped SRAM I/O and verified cycle using **Cadence Xcelium** and analyzed timing behavior via **SimVision** waveform debugging.
- Documented FSM via **waveforms** and **state diagrams**, showing control flow for load, add, and write-back.

Functional Verification & Coverage

- Built UVM-style SystemVerilog environment with driver, monitor, scoreboard, and transactions to validate DUT
- Wrote verification plan with edge-case, overflow, and reset scenarios, using assertions for reset, address, and buffer checks.
- Achieved 99.96% DUT coverage in **Verdi** with constrained-random tests; debugged via **Cadence Verisium** waveforms.

Physical Design Flow & Scripting / STA

- Executed RTL-to-GDSII flow for 32-bit adder in Cadence Innovus: macro placement, clock tree synthesis, and routing.
- Leveraged Linux/Git Bash to edit design files, maintain clean revision histories, and integrate automated TCL scripts
- Built Python scripts to parse timing reports and calculate path delays for timing optimization.

Software Developer AI Trainer (Contract), Atlanta, GA

August 2025 – Present

- Built engineering AI rubrics and prompt templates to evaluate accuracy/formatting using KiCad-based workflow demos.
- Annotated 2,000+ multimodal outputs across classification, ranking, extraction, and error tagging.

Apago, Inc., Electrical Engineering Intern, Alpharetta, GA

May 2025 – August 2025

- Designed control system testbed using oscilloscope, multimeter, and benchtop tools for PCB/SMD soldering, wiring, and diagnostics, which accelerated prototyping at the company by several days and improving system reliability.
- Verified firmware and board bring-up, ensuring correct timing and connectivity between microcontroller and peripherals.
- Helped build and optimized the lead engineer's lab space, designing/implementing makeshift solutions and custom setups to solve hardware gaps, which saves 10+ hours/week in setup, troubleshooting, and workflow interruptions.

Georgia Tech Invention Studio, Intern, Atlanta, GA

June 2024 – July 2024

- Achieved Certified Prototyping Instructor status in under 4 weeks by mastering 10+ fabrication tools (3D printing, resin printing, laser cutting, water jetting, woodworking, metalwork, etc.)
- Used Fusion to design 5+ 3D-printed drone frame iterations, optimizing weight, aerodynamics, and durability
- Designed custom PCBs for flight-control/ telemetry systems using C/C++ on STM32 microcontrollers, leveraging I2C and UART communication for sensor interfacing and motor control

Internationally Genetically Engineered Machine (iGEM), Hardware Research, Suwanee, GA

Nov 2022 – Nov 2024

- Team placed top 10 internationally for developing a low-cost microRNA-based diagnostic tool for coronary artery disease.
- Led development of Micro Q, a \$10 camera-based fluorescence viewer with ~95% accuracy for tracking PCR fluorescence.
- Designed mechanical components in Solidworks; tested and validated system performance in MATLAB.

LEADERSHIP

Hive Makerspace Peer Instructor

August 2025 - Present

- Trained and supported makerspace with 2,000+ members through end-to-end hardware builds, including component selection, schematic interpretation, soldering/rework, and bench debugging, using lab equipment (oscilloscopes, DMMs, power supplies, function generators) while enforcing safety procedures and helping troubleshooting circuit and system-level issues quickly and clearly.

PROJECTS

GT Career Fair Chrome Extension

- Built AI-powered Chrome extension to parse resumes, scrape career fair sites, then rank companies via keyword-matching.

Skills: JavaScript, Chrome Extension, DOM manipulation, Regex, JSON parsing, Web scraping, Algorithm Design

SKILLS

Technical: KiCAD, Verilog/SystemVerilog, C/C++, PCB Design, MATLAB, SPI/I2C/UART Communication, Linux, Git Bash, STM32, Git, Python, Timing Analysis / STA, Microsoft Office Suite,

Relevant Coursework: AI Foundations - Built AI projects in Python using ML fundamentals (SVMs, neural nets, gradient descent), with model testing/validation via NVIDIA AI Makerspace, plus ethical assessment, culminating in an IEEE-style final paper.